

# State of Developer Ecosystem 2024

## Language Trends, Satisfaction, and Strategic Shifts

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Key insights from JetBrains' annual survey of 23,262 developers worldwide covering programming languages, tooling, AI adoption, and career trends.

**23,262**

Developers surveyed

**8 Years**

Longitudinal data

**20+**

Languages tracked

For: Engineering Leaders · CTOs · Developer Experience Teams

# Python Closes the Gap: 8 Years of Language Usage Shifts

% of developers using each language · JetBrains survey 2017–2024

| Language               | 2017 | 2020 | 2022 | 2024 | 8yr Δ | Trend |
|------------------------|------|------|------|------|-------|-------|
| <div></div> JavaScript | 65%  | 70%  | 65%  | 61%  | -4    | ▼     |
| <div></div> Python     | 32%  | 55%  | 53%  | 57%  | +25   | ▲     |
| <div></div> HTML/CSS   | 60%  | 61%  | 54%  | 51%  | -9    | ▼     |
| <div></div> SQL        | 42%  | 56%  | 49%  | 48%  | +6    | —     |
| <div></div> Java       | 47%  | 54%  | 48%  | 46%  | -1    | ▼     |
| <div></div> TypeScript | 12%  | 28%  | 34%  | 37%  | +25   | ▲     |
| <div></div> Go         | 8%   | 19%  | 19%  | 18%  | +10   | ▲     |
| <div></div> Rust       | —    | 7%   | 9%   | 11%  | —     | ▲     |
| <div></div> PHP        | 30%  | 27%  | 20%  | 17%  | -13   | ▼     |
| <div></div> Kotlin     | 2%   | 17%  | 16%  | 14%  | +12   | —     |

• Python +25pp

• TypeScript +25pp

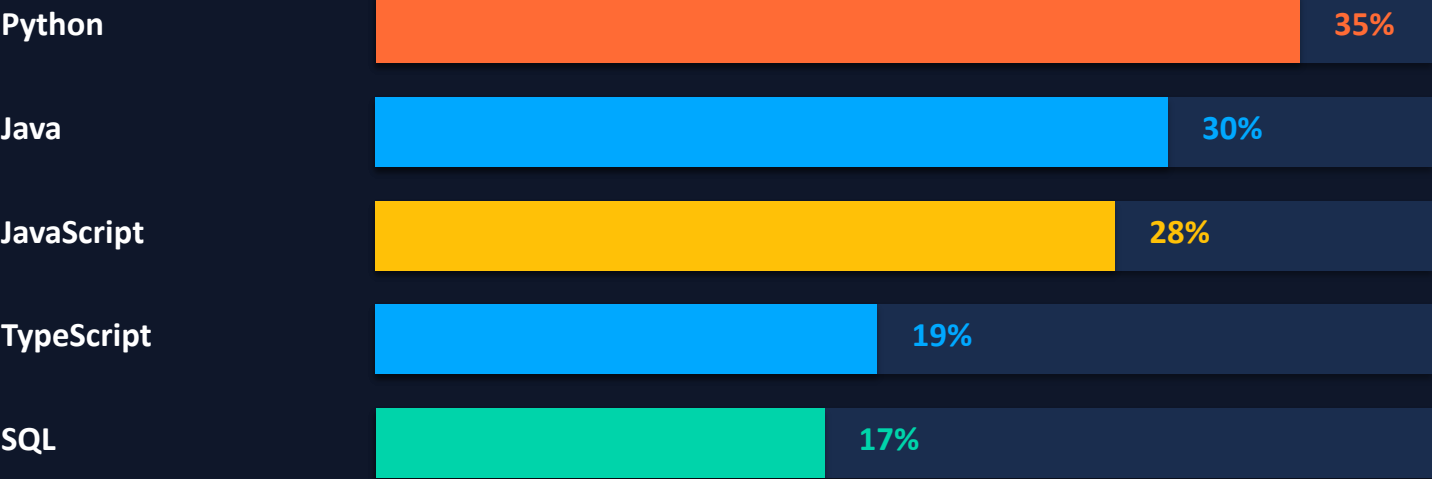
• PHP -13pp

• JavaScript peaked 2020

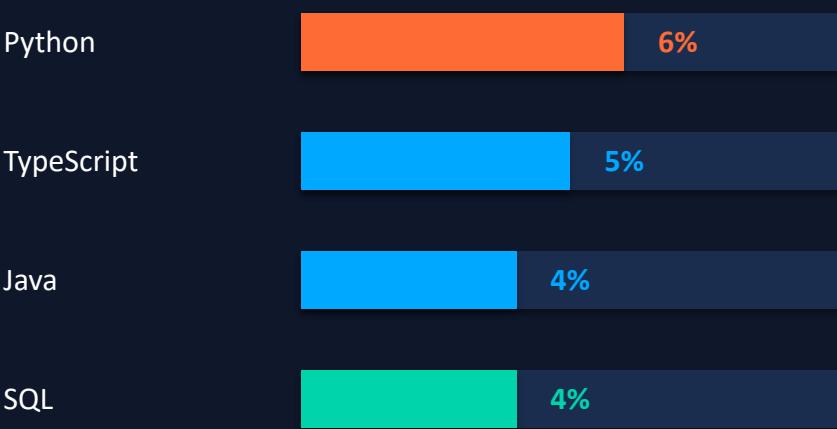
# Python Is Now the #1 Primary Language at 35%

Primary language choice vs. planned adoption intent · 2024

## PRIMARY LANGUAGE SHARE



## ADOPTION INTENT (plan to switch)



### Key insight

Despite strong growth, Python's adoption intent stays at 6% — developers rarely plan to switch primary languages, highlighting deep ecosystem lock-in.

*Note: Adoption intent < 6% across all languages — developers invest deeply in their chosen ecosystem.*

Source: JetBrains State of Developer Ecosystem 2024 · n=23,262 developers

# TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Satisfaction index score (higher = more loved) · top 10 languages · 2024



• TypeScript & Rust: type safety drives love      • Python: breadth + AI ecosystem      • Factor satisfaction into retention strategy

Source: JetBrains State of Developer Ecosystem 2024 · n=23,262 developers

# Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language dominance by application domain · % developers choosing that language for the domain

| Application Domain            | Leading Language | Share | Adoption Bar           | Tier           |
|-------------------------------|------------------|-------|------------------------|----------------|
| AI / Machine Learning         | Python           | 31%   | <div><div></div></div> | Data & AI tier |
| Data Processing & Analytics   | Python           | 17%   | <div><div></div></div> | Data & AI tier |
| Data Science / Analytics      | Python           | 28%   | <div><div></div></div> | Data & AI tier |
| User Interface                | TypeScript       | 62%   | <div><div></div></div> | Frontend tier  |
| User Interface (JS)           | JavaScript       | 58%   | <div><div></div></div> | Frontend tier  |
| APIs & Services               | Java             | 49%   | <div><div></div></div> | Backend tier   |
| Application Logic & Workflows | Java             | 55%   | <div><div></div></div> | Backend tier   |
| Integrating with APIs         | Python           | 38%   | <div><div></div></div> | Backend tier   |
| System Tools & Components     | C++              | 27%   | <div><div></div></div> | Systems tier   |
| Hardware Interfacing          | C++              | 21%   | <div><div></div></div> | Systems tier   |
| Graphics / Rendering          | C++              | 15%   | <div><div></div></div> | Systems tier   |

# What This Means for Your Technology Strategy in 2025

Five actionable recommendations for engineering leaders

01

## Invest in Python for AI/ML teams

Python is the undisputed leader for AI (31%), data science (28%), and data processing. Prioritise Python expertise in data & ML hiring.

02

## TypeScript is the future of frontend & full-stack

TypeScript leads UI development at 62% and holds the #1 satisfaction score (199). Standardise on TypeScript for new web projects.

03

## Rust & Go are the growth plays for infrastructure

Both languages are purpose-built for performance-critical systems and microservices. Rust is still rising; Go has stabilised — both are hiring targets.

04

## PHP & Ruby hiring pools are shrinking

PHP fell from 30% → 17%, Ruby from 10% → 4% over 8 years. Plan migration timelines and account for tighter talent availability.

05

## Type safety correlates with developer satisfaction

TypeScript (199) and Rust (198) lead satisfaction — both enforce type safety. Factor tooling quality & type systems into retention and hiring brand.